

Local Group The small cluster of more than 40 member galaxies to which the Milky Way galaxy belongs. The other major members are the spiral galaxies M31 (the Andromeda Galaxy) and M33. Most of the members are small (or dwarf) elliptical or irregular galaxies. See also *galaxy cluster*.

local sidereal time see *sidereal time*.

luminosity The total amount of energy emitted in one second by a source of radiation, such as the Sun or a star. The luminosity of a star can be expressed in watts or in units of solar luminosity (the luminosity of the Sun is 3.8×10^{26} watts). Stars are divided into luminosity classes denoted by Roman numerals. See also *magnitude*.

lunar eclipse see *eclipse*.

M

MACHO An acronym for MAssive Compact Halo Object, a very low-luminosity object—such as a planet, brown dwarf, exceedingly dim white dwarf, or black hole—that exists in the halo of a galaxy but is usually too faint to be seen directly. MACHOs are believed to account for a relatively small proportion of the unseen dark matter in a galaxy's halo. See also *dark matter*, *halo*.

magnetic field The region of space surrounding a magnetized body within which its magnetic influence affects the motion of an electrically charged particle.

magnetosphere The region of space around a planet within which the motion of charged particles is controlled by the planetary magnetic field rather than the solar wind and the associated interplanetary magnetic field. The shape of a planet's magnetosphere is influenced by the solar wind, which squeezes it inward on the Sun-facing side and drags it out to form an elongated “tail” (a **magnetotail**) on the opposite, or downstream, side. See also *solar wind*.

magnification The increase in the apparent angular size of an object when viewed through an optical instrument, such as a telescope. The magnification of a telescope is equal to the focal length of its objective lens or primary mirror divided by the focal length of its eyepiece.

magnitude (absolute and apparent)

Apparent magnitude is a measure of the apparent brightness of an object as seen in the sky. The fainter the object, the higher the numerical value of its magnitude. The faintest stars visible to the naked eye are of magnitude 6, whereas the brightest objects in the sky have negative apparent magnitudes. A star said to be of 1st magnitude has a magnitude of 1.49 or less, a star of 2nd magnitude has a value of 1.50 to 2.49, and so on. **Absolute magnitude** is the apparent magnitude a star would have if it

were located at a standard distance of 10 parsecs (32.6 light-years) from Earth. See also *luminosity*, *parsec*.

Main Belt see *asteroid*.

main sequence A band that slopes diagonally from the upper left (hot, high-luminosity region) to the lower right (cool, low-luminosity region) of the Hertzsprung–Russell diagram and contains about 90 percent of stars. Main-sequence stars, such as the Sun, shine by converting hydrogen in their cores to helium. See also *dwarf star*, *Hertzsprung–Russell diagram*.

major axis see *ellipse*.

mantle The rocky layer that lies between the core and the crust of a rocky (Earth-like) planet or a major planetary satellite. See also *core*, *crust*.

mare (plural: maria) A relatively smooth, dark, lava-filled basin on the surface of the Moon. The name derives from the Latin for “sea.”

massive compact halo object see *MACHO*.

meridian (1) A great circle on the surface of Earth or another astronomical body that passes through the north and south poles and crosses the equator at right angles. **(2)** A great circle on the celestial sphere that passes through the north and south celestial poles and crosses the celestial equator at right angles. An observer's local meridian passes through the celestial pole, the zenith, and the north and south points of the horizon. See also *celestial sphere*, *great circle*.

Messier catalog A widely used catalog of nebulous objects (most of them nebulae, star clusters, and galaxies) that was published in 1781 by the French astronomer Charles Messier. Objects contained in this catalog are designated by the letter “M” followed by a number. For example, M31 is the Andromeda Galaxy and M42 is the Orion Nebula. See also *New General Catalog*.

meteor The short-lived streak of light seen when a meteoroid plunges into Earth's atmosphere and is heated to incandescence by friction. A **sporadic meteor** is one that appears at a random time from a random direction. A **meteor shower** is a substantial number of meteors that appear to radiate from a common point in the sky (the **radiant**) when Earth is passing through a stream of meteoroids. See also *meteorite*, *meteoroid*.

meteorite A rocky or metallic meteoroid that survives passage through the atmosphere and reaches Earth's surface in one piece or in fragments. See also *meteor*, *meteoroid*.

meteoroid A lump or small particle of rock, metal, or ice orbiting the Sun in interplanetary space. Sizes of meteoroids range from a small fraction of an inch (a fraction of a millimeter) to a few yards (meters). Some are debris from collisions between asteroids. Others are particles released by comets; these spread out along cometary orbits to

form meteoroid streams. See also *asteroid*, *comet*, *meteor*, *meteorite*.

Milky Way (1) The spiral galaxy that contains the Sun, sometimes also referred to as the **Milky Way galaxy** or **the Galaxy**. **(2)** A faint, misty band of light that stretches across the night sky and consists of the combined light of vast numbers of stars and nebulae that lie in the disk and spiral arms of our galaxy. See also *galaxy*.

Mira variable A class of long-period variable stars named after the star Mira—Omicron (o) Ceti—in the constellation Cetus. Mira variables are cool, giant pulsating stars that vary in brightness over periods ranging from 100 days to more than 500 days. See also *variable star*.

molecular cloud A cool, dense cloud of gas and dust in which the temperature is sufficiently low to enable atoms to join together to form molecules such as molecular hydrogen (H₂) or carbon monoxide (CO), and within which conditions are favorable for star formation.

moon Also known as a **natural satellite**, a body that orbits a planet. **The Moon** is Earth's natural satellite. Orbiting Earth at a mean distance of 239,000 miles (384,000 km) in a period of 27.3 days, it has a diameter of 2,159 miles (3,476 km). See also *satellite*.

moon dog See *sun dog*.

multiple star A system consisting of two or more stars bound together by gravity and revolving around each other (a system of just two stars is also called a binary). See also *binary star*.

N

near-Earth asteroid see *asteroid*.

nebula (plural: nebulae) A cloud of gas and dust in interstellar space. The name derives from the Latin for “cloud.” There are several types of luminous nebula (nebulae that shine). An **emission nebula** is a cloud of gas and dust that contains one or more extremely hot, young, high-luminosity stars; ultraviolet light emitted by these stars causes the surrounding gas to glow. Nebulae of this kind are also called HII regions because they contain a large proportion of ionized hydrogen. A **reflection nebula** is observed when the dust particles within a cloud are lit up by light from a neighboring bright star. Other types of luminous nebulae include planetary nebulae (shells of gas puffed out by dying stars) and supernova remnants (the debris of exploded stars). A **dark nebula** (or **absorption nebula**) is a dust-laden cloud that blocks out light from background stars and appears as a dark patch in the sky. See also *diffuse nebula*, *HII region*, *planetary nebula*, *supernova*.

neutrino A fundamental particle of exceedingly low mass, which has zero

electrical charge and which travels at very close to the speed of light.

neutron A particle, composed of three quarks, that has zero electrical charge and a mass fractionally greater than that of a proton. Neutrons are found in the nuclei of atoms. See also *atom*.

neutron star An exceedingly dense, compact star that is composed almost entirely of tightly packed neutrons. A typical neutron star has a diameter of around 6 miles (10 km) yet has about the same mass as the Sun. A neutron star forms when the core of a high-mass star collapses, triggering a supernova explosion. See also *pulsar*, *supernova*.

New General Catalog (NGC)

A catalog of nebulae, clusters, and galaxies that was published in 1888 by the Danish astronomer John L. E. Dreyer. Objects in this catalog are denoted by “NGC” followed by a number. For example, the Andromeda Galaxy is NGC 224. See also *Messier catalog*.

Newton's laws of motion Three laws describing the behavior of moving bodies that were set out by Isaac Newton in 1687. Newton's first law states that a body continues to move in a straight line at a constant speed unless acted on by a force. The second law shows how a force causes a body to accelerate in the direction along which an applied force is acting. The third law states that for any force there is an equal and opposite reaction force.

Newtonian gravity see *gravity*.

nova (plural: novae) A star that suddenly brightens by a factor of thousands or more, then fades back to its original brightness over a period of weeks or months. The flareup occurs when a fusion reaction is triggered on the surface of a white dwarf by gas flowing from a companion star. The name derives from the Latin for “new,” because the rapid brightening produces what appears to be a new star. See also *white dwarf*, *fusion*.

nuclear bulge see *spiral galaxy*.

nuclear fusion see *fusion*.

nucleus (plural: nuclei) (1) The compact central core of an atom, which consists of a number of positively charged protons and neutral neutrons. The nucleus of a hydrogen atom consists of a single proton. **(2)** The solid, ice-rich body of a comet. **(3)** The central core of a galaxy, within which stars are relatively densely packed together.

O

occultation The passage of one body in front of another, which causes the more distant one to be wholly or partially hidden. The term is usually used to describe the passage of a body of larger apparent size in front of a body of smaller apparent size—